

SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID – SIH26097
- Problem Statement Title – AI-driven voice assistant for livelihood mapping and NSQF-aligned skilling recommendations for SC communities under the GIA component of PM-AJAY
- Theme – Agriculture, FoodTech & Rural Development
- PS Category – Software
- Team ID – _____ (fill from the SIH portal)
- Team Name (Registered on portal) – Cognify



SMART INDIA HACKATHON 2026

Proposed solution · one free phone call that turns a person's skills into the right work



The problem SIH26097 · Ministry of Social Justice & Empowerment

PM-AJAY pays for skill training and small-business support (up to ₹50,000) for poor SC families. But people are picked and placed through paper forms and field visits. Those who can't read, or don't know the trades, land in the wrong course — and don't get a job.

41%

of trained people got a job

Source: CAG audit 2025

43%

of rural women have no phone of their own

Source: NFHS-6, 2023-24

34%

of SC adults cannot read

Source: Census 2011

WHO WE SERVE 20.14 crore SC people → ≈ 5.9 crore below the poverty line (eligible) → ≈ 50,000 get GIA support each year → < 1 in 1,000 a year → every seat must fit

Census 2011 (via MoSJE Handbook 2021) · SC poverty ratios 31.5% rural / 21.7% urban (Planning Commission, 2011-12) · GIA: 2,01,006 beneficiaries in 2021-25 → ≈ 50,000 a year (PIB, 19 Mar 2025)

HOW IT WORKS — ONE CALL, ABOUT 10 MINUTES, ON ANY PHONE



1

Missed call → call-back

He gives a missed call; SkillCall calls back in a minute — free, from a govt 1600 number, in his language.



2

Facts by keypad

Age, education, travel distance: he presses keys. A speech error can never change eligibility.



3

He talks, AI listens

"Aap kya kaam karte ho?" The AI finds his official job (NCO code) and asks about his skills.



4

Read back & saved

Everything is read back; he presses 1 to confirm. Saved as a clean profile — no paper form.



5

1-4 options that fit

Only courses, schemes or jobs near him that truly fit. Every fact checked. He can ask any doubt.



6

Apply & follow-up

SMS with Case ID + nearest CSC; officer approves in the AJAY app; calls at 1, 3 and 6 months.

WHAT MAKES IT DIFFERENT



Smart questions

Next question depends on his last answer: bike mechanic → engine, tyres, oil.



Profile writes itself

AI fills the full profile table in the proper format. Officers only review and approve.



Only verified govt data

Answers come only from loaded data: rules, NQR courses, NCO codes, centres. Nothing invented.



SATYA fact-check

Each fact is matched to a database row before it is spoken. No match → not said.



Cheapest AI first

Small model by default, bigger only if needed, a human if still stuck. Common answers get cached — cost falls over time.



Web portal too

Mobile + OTP login: family, CSC or officer can view or edit the case — or skip the call.

Also answers the 5 GIA issues in the brief: planning · beneficiary selection · placement after training · coordination · too few ground staff

TECH STACK

Calls & voice

Exotel / Plivo

WhatsApp API

Pipecat voice server

AI models

Saaras v3 • STT

Bulbul v3 • TTS

Sarvam-30B / 105B

IndicConformer

Bhashini

SATYA (ours)

Backend & data

Python

FastAPI

Celery

Redis

PostgreSQL

pgvector

Apps & govt links

React

Leaflet maps

NCO / NQR

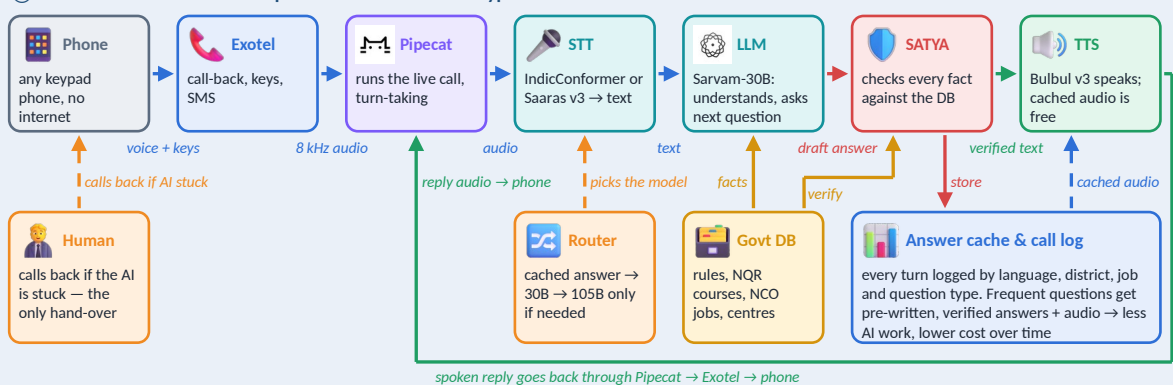
AJAY app*

DigiLocker*

Docker • MeghRaj*

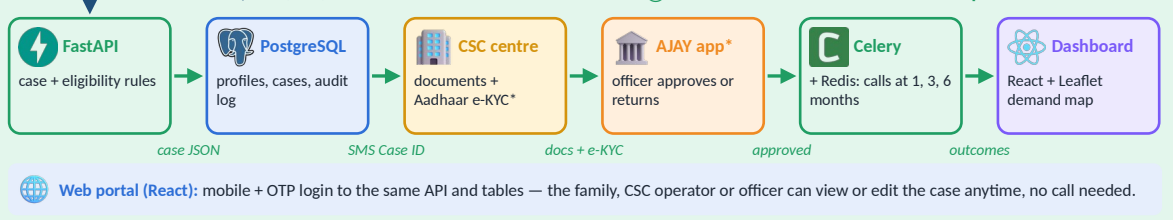
HOW IT WORKS — ARCHITECTURE

① DURING THE CALL — each spoken turn = 1.2-2.2 s • keypad turns are instant



call ends → confirmed profile (JSON) → FastAPI

② AFTER THE CALL — the confirmed profile becomes a case



AI understands → verified data decides → SATYA checks → humans approve

* needs a government MoU.

[Full architecture, API docs and code → GitHub repo \(add link\)](#)

Data rules: no Aadhaar number stored · raw audio deleted after confirmation · consent logged (DPDP Rules 2025) · every edit in the audit log

Can it be built and run? Yes — every part already runs in India, the prototype can be built today, and one completed call costs about ₹25–54.

ALREADY WORKING IN INDIA — NOTHING HERE IS UNTESTED



10 languages

UIDAI already runs Sarvam's voice AI for Aadhaar on its own servers, in 10 of Sarvam's 22 languages.

Source: PIB / MeitY 2025



12 lakh calls a day

Kilkari, the govt health-advice line, already handles this volume every day.

Source: BMJ Global Health 2021



≤ 20% word error

Best Indian speech AI on real phone calls — 36,691 speakers, 15 languages, 536 hours.

Source: Voice of India 2026 (arXiv)



1.2–2.2 s per reply

End of speech 0.5 s + speech-to-text 0.1–0.3 s + AI first word 0.3–1 s + voice 0.2 s + network 0.1–0.2 s.

Source: Sarvam docs • NVIDIA 2026

WHAT ONE COMPLETED CALL COSTS — ₹ PER PERSON

Completed call = 10-min interview + options, plus 3 follow-up calls of 2 min (= 16 min of phone time).

Cost item	How we calculate it	Low	High
Phone line (Exotel)	16 min × ₹0.80–1.80, + unfinished calls & 30-s billing	₹14.7	₹33.7
Speech → text (Saaras v3)	₹30 per hour; only his speech (50–60% of the call)	₹4.2	₹5.1
Text → speech (Bulbul v3)	₹3 per 1,000 characters; fixed questions pre-recorded	₹2.4	₹6.0
AI brain (Sarvam-30B / 105B)	~90% of turns on 30B; 105B only when needed	₹0.25	₹0.5
SMS + human call-back	4–6 SMS; 10–20% of callers need a human for 5–6 min	₹2.2	₹5.3
Hosting & helpline number	servers, database, number — spread per call	₹1.0	₹3.0
TOTAL per completed profile	1,000 missed calls → 420–517 finished profiles	≈ ₹25	≈ ₹54

Falls further over time: frequent questions get pre-written, verified answers and pre-recorded audio (answer cache, slide 3), and Sarvam bills cached input at ₹10.98 vs ₹29.28 per 1M tokens. At national scale ~70% of speech moves to free self-hosted IndicConformer → ₹21–49 per profile.

Prices: Sarvam API pricing • Indian telephony rate cards 2026 • funnel = our estimate

RISKS → HOW WE HANDLE THEM

Village dialects are hard → keypad for facts • read-back • bigger model • then a human

AI could say something wrong → SATYA blocks anything not in the govt database

People ignore unknown numbers → he calls first; call-back from an official 1600 number

Slow replies bore callers → streaming voice • pre-recorded questions • keypad turns are instant

WHO GETS THROUGH — AND WHO NEEDS A HUMAN

- 50% answer on the 1st attempt, 76% by the 3rd, 99.5% by the 9th
- ~80% of answered calls finish the interview; ~85% of those finish the options → 420–517 profiles per 1,000 missed calls
- 10–20% need a human counsellor call-back — design target ≤ 20%
- Can't be reached by phone? The web portal (mobile + OTP) is always open.

Source: Kilkari, BMJ Global Health 2021

Source: our funnel estimate

Source: our design target

CAN THE GOVERNMENT AFFORD IT?



0.05–0.10% of the budget All of India: ₹1.1–2.1 crore a year vs PM-AJAY's ₹2,140 crore (2026–27).



Under ₹1 in every ₹100 ≈ ₹220–420 per beneficiary vs = ₹60,700 average GIA support (₹1,219.80 cr ÷ 2,01,006).



₹9–13 lakh to start 6-month pilot in one block (≈ 1,000 families): build, test with real users, then scale.

Who gains? SC families, CSC operators, district officers and the Ministry. Every number shows how it was worked out.

WHAT CHANGES



More trained people get jobs

41% → 70%

Only 41% got a job; 70% is the govt's own target. Per 1,000 trainees, even 50% means 90 more people working.

Source: CAG 2025 · MSDE norms



More people stay in the job

62% → 73%

Trainees told the real pay and place before joining were 11 percentage points more likely to stay 5+ months (men). SkillCall tells him on the call.

Source: J-PAL · Chakravorty et al. 2024



Officers get their time back

30 h → 5 h a month

Per officer, 40 cases a month: today 45 min each (interview + 3 follow-up calls) = 30 h; with SkillCall a 7.5-min review = 5 h. India: 50,000 cases × 37.5 min = 31,250 h a year = 18 full-time officers.

Our estimate: AJAY app's data entry not counted



Less money wasted

₹12.5–15 cr a year

If wrong matches fall in 5 of every 100 cases: $2,500 \times ₹50,000 - 60,700$ support = 6–14× SkillCall's yearly cost.

Our estimate: PIB 2025 average support

REACH SkillCall can profile 1–1.5 lakh people a year (2–3 applicants screened per GIA seat), so each of the = 50,000 yearly seats goes to the person it fits best.

WHO WINS



SC family

Right course the first time, free, in his own language — and calls until he is earning.



CSC operator

The case arrives pre-filled; only documents and e-KYC are left to do.



District officer

Ready, checked cases in the AJAY app; a demand map by block for the yearly plan.



Ministry

One record from call to job; each outcome marked confirmed (employer / EPFO*) or self-reported.

BENEFITS



Social Women and non-readers included; right course the first time.



Economic Less money on wrong training; more people earning — proved by follow-up.



Governance One shared case for CSC, officer, partner and bank; planning data, not paper.



Environment No paper forms, fewer trips to offices, runs on the shared govt cloud.



Government data

PIB, Rajya Sabha reply (19 Mar 2025) — GIA: 2,01,006 beneficiaries, ₹1,219.80 cr pib.gov.in

PIB PM-AJAY explainer (May 2025) — eligibility, 5% admin & MIS line pib.gov.in

CAG Report No. 20 of 2025 — PMKVY: 41% placed, no skill-gap analysis cag.gov.in

NFHS-6 (2023-24) — women's phone ownership 63.6%, rural 57.4% rchiips.org

Census 2011 (MoSJE Handbook 2021) · SC poverty ratios 2011-12 (Lok Sabha) socialjustice.gov.in



Technology

Sarvam API pricing & docs — Saaras v3, Bulbul v3, 30B / 105B; voice-agent guide docs.sarvam.ai

NVIDIA × Sarvam 2026 — Sarvam-30B serving, first token < 1 s (P95) developer.nvidia.com

Artificial Analysis — Sarvam-30B ≈ \$0.03 per 1M tokens (blended) artificialanalysis.ai

AI4Bharat IndicConformer · Pipecat · Bhashini ai4bharat.iitm.ac.in



Research

Chakravorty et al. 2024 / J-PAL — honest job info: 62% → 73% stay in job povertyactionlab.org

Bashingwa et al. 2021, BMJ Global Health — Kilkari call answer rates gh.bmj.com

Voice of India 2026 — speech-AI benchmark on real Indian phone calls arxiv.org

Medhi et al. 2011, ACM TOCHI — voice interfaces for low-literacy users dl.acm.org



Standards & policy

PM-AJAY portal & AJAY app launch (26 May 2026) — PIB pib.gov.in

NCO-2015 occupation codes · NQR / NCVET job roles nqr.gov.in

DPDP Rules 2025 — consent and data protection meity.gov.in

TRAI 1600-series rules for government calls (2026) traai.gov.in

MeitY (Lok Sabha 2023) — 5.58 lakh CSCs, 4.44 lakh at gram panchayats eparlib.nic.in

SEE MORE — EVERYTHING THE SIX SLIDES COULD NOT FIT



GitHub repo

code, architecture docs, API reference

[\(add link\) →](#)

Live prototype

try a SkillCall conversation in the browser

[\(add link\) →](#)

Demo video

2-minute walkthrough of a real call

[\(add link\) →](#)

Calculations sheet

every number, its source and full working

[\(add link\) →](#)